

Anthony Chen

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EDUCATION

University of Washington

Master's, Technology Innovation (Robotics Track)

09/2025 - 04/2027

GPA: 3.94

- Award: MSTI Merit Scholarship Awardee
- Interdisciplinary master's program merging engineering and business with a focus on product prototyping and innovation

University of California - Los Angeles

Bachelor's, Computer Science

09/2021 - 06/2025

GPA: 3.8

- Relevant Coursework: Software Construction, Computer Architecture and Organization, Data Structures & Algorithms, Discrete Structures, Operating Systems, Computer Graphics, Programming Languages, Machine Learning, Artificial Intelligence, Natural Language Processing, Computer Vision

SKILLS

Skills: Javascript, TypeScript, Python, C/C++, HTML/CSS, React.js, Next.js, Express.js, Node.js, Git, GitHub, Docker, Firebase, Postgres, MongoDB, SQL, NoSQL, Linux/Unix, REST APIs, AWS, Pytorch, ROS2, SLAM, Nav2, Gazebo, RViz, OpenCV, YOLO, CircuitPython, ESP32, Motor Control, I2C, PID Control

PROFESSIONAL EXPERIENCE

Ellison Medical Institute

Software Engineering Intern

Los Angeles, CA, USA

06/2023 - 09/2024

- Developed and deployed a containerized web dashboard (Next.js, TypeScript, Docker, Nginx) for computer vision models, modernizing internal workflows and integrating with internal servers for secure access
- Designed a scalable database schema and implemented the app architecture for a project lifecycle tracking tool, enabling the monitoring of 60+ research projects and new proposals, with support for custom approval workflows and future scalability
- Collaborated across cross-functional teams to build a Python data pipeline that automated the upload of 500+ patient medical records via REDCapCloud's REST API, ensuring data integrity through automated testing, audit logs, and backups

UCLA Samueli School of Engineering

Engineering 1 Instructor - Autonomous Rover

Los Angeles, CA, USA

09/2023 - 06/2025

- Instructed 100+ students in designing and programming autonomous rovers to navigate an obstacle course and secure objects
- Presented lectures on essential concepts including SolidWorks, C++, Arduino programming, and circuiting/wiring

PROJECTS

Autonomous Cinematography Robot

Robotics Software Engineer

Seattle, WA, USA

01/2026 - 03/2026

- Architected a multi-node ROS2 system on Linux/Docker for a customized TurtleBot3, integrating YOLO object detection with a custom controller to perform real-time subject centering and navigation
- Implemented a PID camera servo controller fusing camera bounding box offsets with LiDAR data to minimize tracking error and enforce obstacle avoidance constraints
- Integrated Nav2 and SLAM for dynamic obstacle replanning, iterating through Gazebo simulation to validate navigation and tracking behavior before hardware deployment on the physical TurtleBot3

Autonomous Maze Navigation & Marker Localization - TurtleBot4

Robotics Software Engineer

Seattle, WA, USA

01/2026 - 03/2026

- Built a custom ROS2 package on Linux/Docker implementing a state machine to orchestrate autonomous maze traversal, ArUco marker localization, and precision docking, achieving top placement in a class-wide timed competition
- Integrated SLAM Toolbox with Nav2 for real-time occupancy grid mapping and autonomous goal navigation, validating system behavior in Gazebo before physical deployment
- Developed an OpenCV pipeline to detect and localize ArUco fiducial markers, fusing pose estimates with the SLAM map to dynamically update navigation goals

Human-in-the-Loop Robotic Vehicle Control System

Embedded Software Engineer

Seattle, WA, USA

09/2025 - 12/2025

- Engineered a dual-microcontroller embedded system (ESP32-C3/S3) with BLE UART for real-time bidirectional wireless communication between a custom handheld controller and autonomous robot vehicle
- Implemented ADXL345 accelerometer-based fall detection with zero-offset calibration and configurable Z-axis threshold filtering for reliable event detection on resource-constrained hardware
- Designed a finite state machine for robot behavior control with PWM-based L298N motor control, difficulty-scaled speed profiles, and interrupt-driven human override for real-time corrective steering

Autocreator

Backend Engineer

Remote

08/2024 - 12/2024

- Collaborated as part of a 3-developer team to create a full-stack web application designed to streamline social media content generation and scheduling, improving workflow efficiency for users
- Architected a scalable microservices backend with Express/Node.js and containerized Python workers using AWS Lambda, SQS, and S3 to manage asynchronous content generation efficiently
- Implemented user authentication and social media connection workflows using OAuth APIs from TikTok, Instagram, and YouTube to enable secure user login and account linking

Leo Fitness | UCLA Creative Labs

Backend Engineer

Los Angeles, CA, USA

01/2024 - 03/2024

- Architected RESTful APIs to enable CRUD operations for a workout plans and activities tracking app built with React Native, MongoDB, Express/Node.js
- Developed and deployed MongoDB database and backend Express/Node.js server infrastructure on AWS Elastic Beanstalk
- Automated API documentation using SwaggerUI, streamlining collaboration between teams for efficient testing and integration